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Vehicle Monitoring for Traffic Violation Using V2I Communication

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This paper presents a way of using the Can bus to connect different types of sensors and ZigBee as the technology to c

***Abstract*— The fundamental concept of connectivity is IoT which can be integrated into traditional communication network to solve many problems. Vehicle-to-Infrastructure (V2I) technology is an aspect of IoT enabling Intelligent Transport Systems. Vehicular monitoring as part of V2I helps decrease the problem caused by vehicles in the city, like congestion, traffic violation and road accidents. To regulate these, an instance of Vehicle-to-infrastructure communication model realizing data transmission between traffic light and vehicle is observed in this paper. A prototype of On-Board Units (OBU) and Road Side Units (RSU) are developed. Vehicles send speed, identity and location messages to the traffic controller based on ZigBee wireless technology. The message is analyzed to check speed violation. Information related to user's driving is transmitted to a monitoring server from the RSU to charge the offender of Traffic rules. Therefore, modules based on Controller Area Network(CAN) bus for in-vehicle communication, OBU details for Over speeding in highways, RSU information for Real time data collection and E-mail notification service based on violation rules are developed.**

Keywords: V2I, CAN bus, Speed violation, Violation Monitoring Server.

I. INTRODUCTION

IoT is the state of art technology used in many applications like smart transport system, smart homes, smart city design etc. IoT is popularized due to increasing internet users and devices that can be connected to the internet which allows these devices to acquire data and communicate with each other, that makes them smart. The world gets smarter as more devices gets connectivity. IoT is one of the key factor for the emerging technology- "connected cars" that interconnects itself to other devices including other cars, infrastructures, home, offices etc. With the rise in number of vehicles, it has become of great importance to develop an Intelligent Transport System (ITS) that could reduce the road accidents and traffic congestions.

Traffic violation has become common leading to numerous road accidents every year. India witnesses 17 deaths and 55 road accidents every hour which is approximately 480,652 lakhs in 2016 [1]. One of the main reasons for high number of fatalities is reckless driving. Vehicular Ad-hoc Networks(VANETS) and ITS can reduce the number of fatalities. To have such secure traffic regulatory system [2] it would require the information to be shared between vehicles and infrastructure with minimal human intervention. Therefore, real-time road traffic information is a key aspect to build a system that improves travel comfort and provides protection.

V2I communication is the exchange of data between vehicles and the infrastructure that are enabled by a system of firmware, hardware and software helps in capturing data and share timely information to enable connectivity. The computing devices are embedded into vehicles and infrastructures to **digitalize** the V2I system. The vehicles communicate with infrastructures through wireless communication network. Communication protocols like WLAN, ZigBee, DSRC, WAVE etc. are used for V2I communication. These information can be used for services like parking slot booking, traffic jams notifications, emergency vehicle routing etc.

The vehicles and the infrastructure are within a network, hidden under a firewall and can be remotely monitored by connecting to cloud services. These features could be meshed in an architecture, offering a new approach for handling traffic violations and monitoring. This work deals with development of system for tracking of vehicles that violated traffic rules using V2I communication. The system employs combination of Road Side Units(RSU) and On-Board Units(OBU) which collects the violated vehicle details in real-time and feeds to a control center (Monitoring server) in order to charge the offender.

The paper is structured as: Related works and motivation in context of V2I communication in Section II. System Overview in Section III. Section IV gives description of system implementation. The results and experiments are discussed in Section V and final section addresses the issues of future works to improve the traffic violation management.

II. RELATED WORKS

The European Transport policy [3] uses intelligent transport system as a key for improving safety and traffic monitoring. Vehicles rely on various communication networks to transmit data to different entities called the Road Side Units. Data is processed and translated into useful information helping the authorities to identify the violated vehicle. In a V2I communication, vehicles communicate to RSU or remote-

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control center by using wide area networks. The infrastructure serves by gathering real time information and "suggest" behaviors to the drivers or the managing authority. For example, an Audi car; the first commercial V2I system enabled car in U.S. which communicates with traffic controller to know the duration of red light. Such systems help save fuel and provide commercial information. Periodic messages are sent to know vehicle's status like position, speed and identity. [4].

The IEEE (Institute of Electrical and Electronics Engineering) and ASTM (American Society for Testing and Materials) acquires DSRC (Dedicated Short-Range Communication) standard providing wireless communication for applications within 1000m range in a speeding highway [5]. Wireless communication to Vehicle is established using WAVE (Wireless Access vehicular Networks). The challenge for this type of communication is to cope with the fast-changing network with reliability and timing. [6]. Therefore, identification of vehicle becomes a problem. RFID technology is used to collect data from the vehicle and deliver it while crossing any requested road side unit. But this system requires additional hardware other than the OBU [7]. The solution is to use a unique ID assigned to the content of message frame instead of a device. This method is known as Named Data Network(NDN). Vehicular NDN is a request reply communication model. The data is more secure than IP based communication. [8]

A wireless standard is suggested by [9]: ZigBee. It was chosen for V2I because of its "strong self-organizing, strong self-healing capabilities, big network capacity and good communication reliability." The vehicle violation is detected by interfacing GPS and speed sensor which is monitored by GPS signal and accelerometer of the car in on-board system, the location and maximum speed is sent to the authorized personnel using GPRS message that issues violation ticket to the driver [10]. Wireless communication technologies such as 3G,4G, LTE, IEEE 802.11, IEEE 802.16e are used for long range connectivity which are effective for V2I communication. Therefore, wireless access point are the key components of V2I communication [11]. The police officers are given a hand-held device that are notified whenever a violation is registered. It reads the RF tags installed in the vehicle's number plate which will also contain the vehicle ownership data and the violation penalty to the driver [12]. Traffic violation and ticketing system architecture considers content-centric and vehicular networking which detects the offenders and issue the ticket to the violator without human intervention [13]. Various studies have proposed idea related to this paper. A Hadoop-based solution is proposed for vehicle data that processes RPM, speed, GPS location etc., which are obtained by On-Board Diagnostics-Bluetooth scan tool. An Android application is developed so that users can see the diagnostic information of the vehicle stored in a database on cloud [14].

The idea of this paper is to rely on messages received from On-Board Unit data to the Road Side Unit that will diagnose for traffic violations when RSU connects to a central server system using Wi-Fi and mail the authority in-charge so that necessary actions can be taken on the violator.

III SYSTEM OVERVIEW

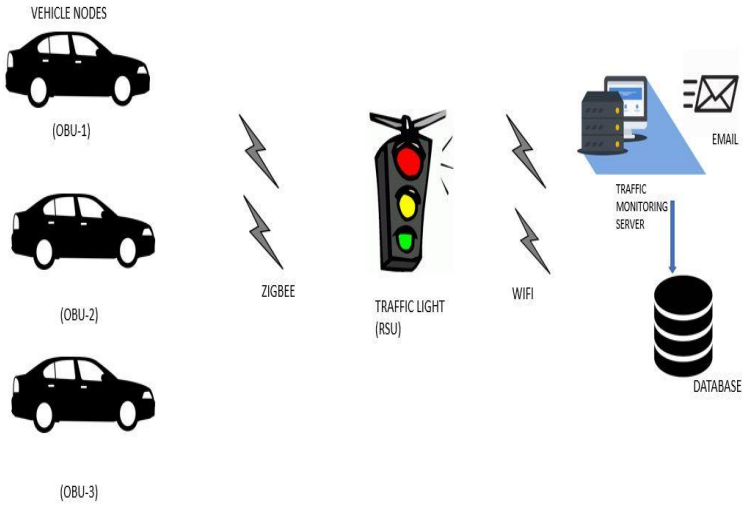


Fig. 1. System Design

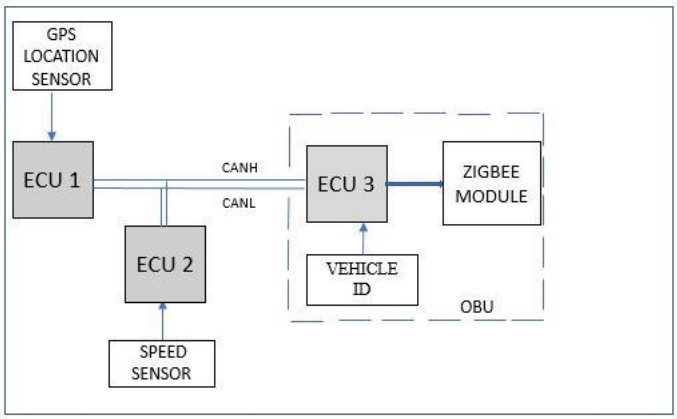


Fig. 2. Vehicle Unit

A. General Overview

The overview of this system is shown in Figure 1. The system consists of three modules namely the Vehicle Unit, the Road Side Unit and the Monitoring Server. The vehicle unit is a device installed in the vehicle to collect information like speed, position data(latitude/longitude) and vehicle ID. The data is collected and ready to be transmitted to a Road Side Unit once the vehicle is in range. The vehicles continuously broadcast the information to the RSU. Similarly, the road side unit may gather information that are localized to vehicles in the specific area. Road Side Unit is connected to a database and monitoring server through a backbone network of Wi-Fi.

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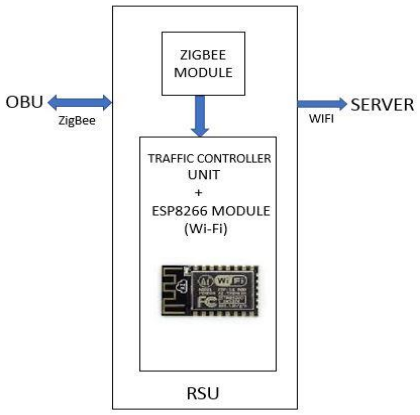


Fig.3. Design of RSU

B. Vehicle Unit

The Vehicle Unit collects information from vehicle as shown in Figure 2. It consists of an ECU (Electronic Control Unit) which collects the sensor data from like speed, GPS location and vehicle ID. To avoid redundant wiring, CAN bus is used as the standard for network communication between ECUs. The data from ECU1 is transmitted using CAN bus to the ECU 2. On Board Unit consist of ECU-2 and the ZigBee module to communicate with RSU. There are three interfaces I/O, CAN bus and Serial interface. I/O interface to connect to sensors, CAN interface between the ECUs and serial interface to connect to ZigBee module.

C. Road Side Unit

The proposed Road Side Unit routes information from vehicle unit to the monitoring server. It is composed of a ZigBee module, a microcontroller with integrated Wi-Fi module as shown in Figure 3. The ZigBee module communicates with the OBU when the vehicle is in range. The Road Side Unit identifies the vehicle details received from the vehicle unit. The ID helps in distinguishing it. Therefore, RSU does not depend on any sensor as it relies only on the message frame received from the OBU to identify the violated vehicle.

D. Monitoring Server

The traffic monitoring server and the database is wirelessly connected using Wi-Fi protocol (IEEE 802.11 standard) from the Road Side Unit. Database is used to store the information from the vehicles and the RSU. When violation of a vehicle is inferred, the monitoring server sends E-mail to the authority in-charge with the vehicle details fetched from the database. SMTP (Simple Mail Transfer Protocol) is an application layer push protocol used for sending email. The process of SMTP is called “store and forward”. The protocol helps the client(RSU) in opening a TCP connection on the SMTP server. On listening to response, SMTP initiates a connection on that port which helps the client to send mail instantly.

IV. SYSTEM IMPLEMETATION

A. System Prototype

A prototype of OBU and RSU are implemented using ARM mbed LPC 1768 (Cortex M3) microcontroller and Node Mcu with ZigBee module for wireless connectivity between OBU and RSU. The RSU consists of ESP8266 based microcontroller (NodeMcu) as the traffic light controller and a ZigBee. On the server side, MySQL database is used to store the information hosting on Apache Server

.

B. Data Collecting Unit

The vehicle unit senses speed and location(GPS) information that are interfaced to ECU1 from its respective sensors. This information also contains unique ID of the vehicle which is sent through CAN bus to ECU 2. The ZigBee module sends the Vehicle details to the RSU when the receiver module is in range.

C. Data Format

A custom data format for communication between OBU and RSU was created as shown in figure 4. A message frame has 12 bytes, first byte denotes the start of frame (SOF) and uses a fixed value, say 0x01 which signifies the start of message. The next two bytes are the vehicle ID. Then is, information of the vehicle. Two bytes(2B) for speed, Four bytes(4B) for latitude, Four bytes(4B) for longitude. The speed collected by OBU is an integer, GPS location are floating point values with maximum 3 decimal places. The last one byte(1B) is for End of Frame (EOF) which is again a fixed value denoting the end of datum.

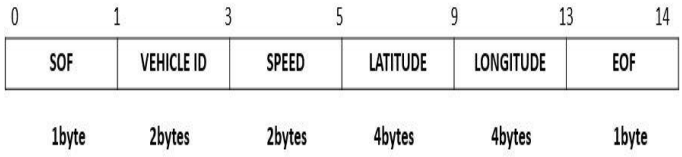


Fig 4. Frame Format

The OBU sends message in the above frame format, and when the traffic light controller receives it, the data is decoded and parsed in the same manner to get the message.

D. Traffic Controller Module:

This module is an important part of the system. The vehicle information from OBU is received by the Road Side Unit. When the traffic signal is Red, RSU receives the vehicle ID, checks for next bit, the speed of vehicle. The controller compares to a speed threshold by calculating the RPM of the wheel. When the rpm is more than 180 a flag is raised on traffic violation, speed and GPS location with vehicle details are stored in a database using Wi-Fi. The information is timestamped in it. A mail is triggered from the monitoring server to the authority in charge with the information stored in database.

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V.EXPERIMENTS & RESULTS

The hardware implementation for traffic light monitoring is as follows.

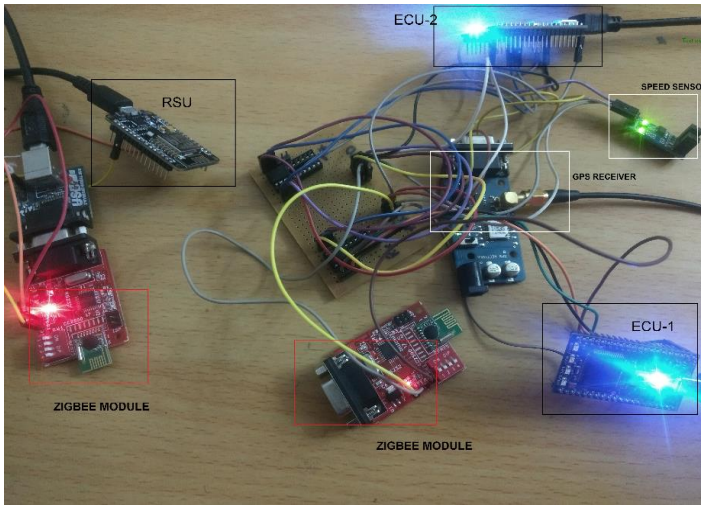


Figure 5. Hardware Implementation

Figure 5 presents the hardware setup. The following Figure 6 shows the output for communication between OBUs. and RSU. Once the vehicle crosses the RSU, it gets connected using Wi-Fi as shown and the vehicle details are obtained. The controller checks for the violated vehicle data and its pushed into DB as shown in Figure 7. The ID maps to another table containing address and details of the violator. An email notification is forwarded to the authority whenever a traffic violation is detected. This is implemented by SMTP mailing protocol by establishing a session between the monitoring server and SMTP server.

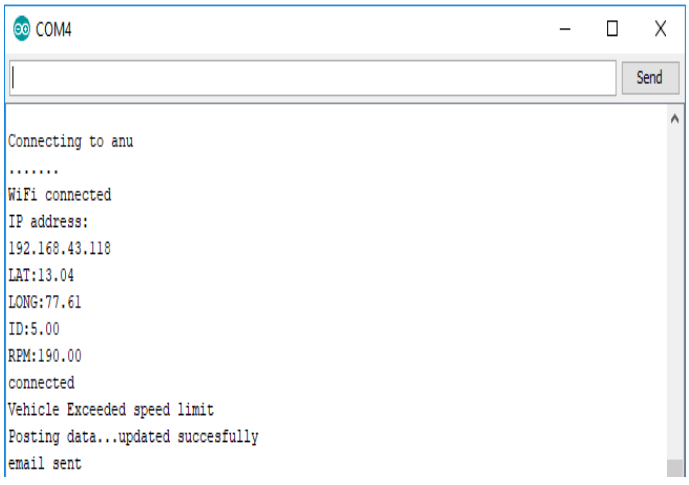


Fig 6. RSU collecting OBU details

The database is updated with speed and location details that are time stamped as seen in Figure 6. XAMPP, a cross platform web server, My SQL DB is used and hosts on Apache server as seen. The scripts are written in PHP language. Figure 8 shows

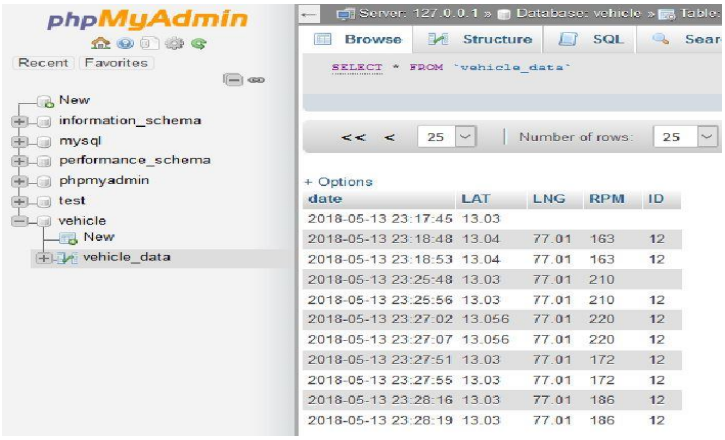


Fig. 7. OBU details stored in DB

mail notification on event of traffic violation with the OBU details sent to the authority in charge.

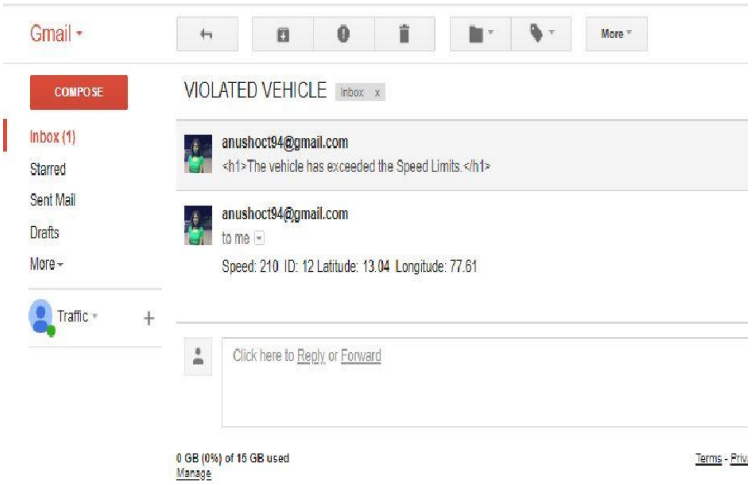


Fig 8. Mail trigger on speed violation

VI. CONCLUSION

Traffic Violations and road accidents are one of the most serious problems today. This paper proposes a V2I based communication to monitor the traffic violations. The prototype is developed based on CAN bus and IEEE 802.15.4 to obtain and transmit vehicle information to the infrastructures. SQL based database was used for storing the vehicle diagnostics. This system will identify the drivers violating the rules without any effort for the authority to chase and charge them. The RSU sends an email to the authority. As a result, manual operations and delays are reduced. Although the RSU depends only on the message frame for identifying the vehicle, it is difficult to determine the best scheme for communication since highly sensitive data is shared. Therefore, the future works could be based on improvement of communication protocol, security and privacy issues during data transmission.

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